School of Future Tech

Case Study Report

on

Project Manager

by

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1. Introduction to the Case Study

In the modern digital landscape, individuals manage numerous accounts across various platforms, including email, social media, and financial services. Remembering unique, complex passwords for each account is a significant challenge. This case study focuses on the design and implementation of a **Secure Password Manager** using C++. The tool provides a centralized, encrypted "Digital Vault" to store, retrieve, and manage credentials effectively via a Command-Line Interface (CLI).

The project demonstrates fundamental software engineering principles such as data security through encryption, persistent storage using file handling, and user-centric design for managing sensitive information.

2. Problem Statement / Case Background (Abstract)

Background

Most users resort to using simple, identical, or easily guessable passwords for multiple accounts, which significantly increases the risk of unauthorized access. Physical records or plain text files are inherently insecure. Rule-based or manual management is no longer sufficient for maintaining high security across dozens of digital identities.

Abstract

This case study presents a C++ based solution for personal credential management. A secure vault is implemented where passwords are not stored in plain text but are protected using a **Shift Cipher (Shift +3)** algorithm. The system allows users to securely save site-specific usernames and passwords into a persistent file vault.txt. It includes features for searching and decrypting specific records and deleting obsolete ones. Evaluation shows that this approach effectively protects data at rest while providing a simple, efficient interface for the user.

3. Case Study Design

The Password Manager is designed as a modular system with a clear data flow pipeline:

1. **User Input Stage**
    - Collection of Website Name, Username, and Password from the user.
2. **Encryption Module**
    - Transformation of the plain-text password into a ciphered format using the Shift algorithm.
3. **Data Storage Module**
    - Appending the encrypted record to vault.txt using a pipe delimiter.
4. **Retrieval & Search Module**
    - Scanning the vault file for a specific site name provided by the user
5. **Decryption Module**
    - Reversing the shift algorithm (Shift -3) to display the original password to the authorized user.

6. **Maintenance Module**

    - **Logic** to filter and remove specific records by rewriting the vault file accurately

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key methods and algorithms:

1. **Programming Language: C+**

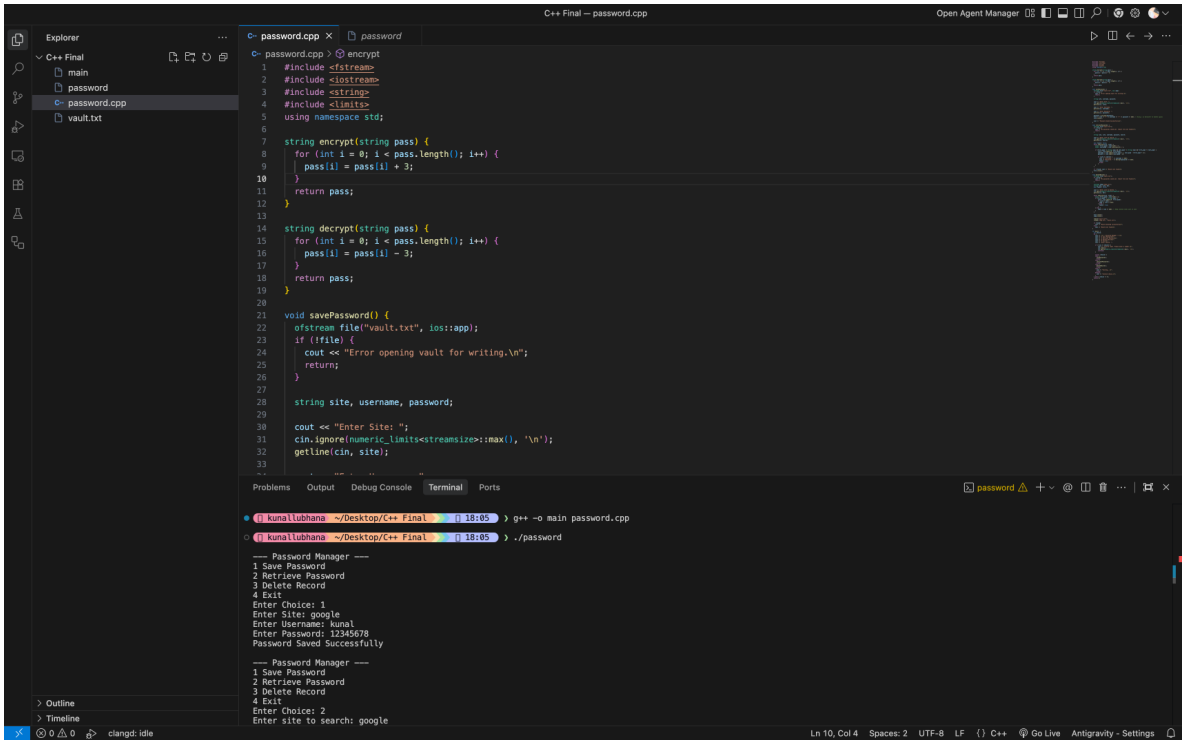
2. **Libraries**
    - fStream for For persistent storage in text files
    - iostream for standard input and output
    - String for For text manipulation.
    - Limits For robust input buffer management.

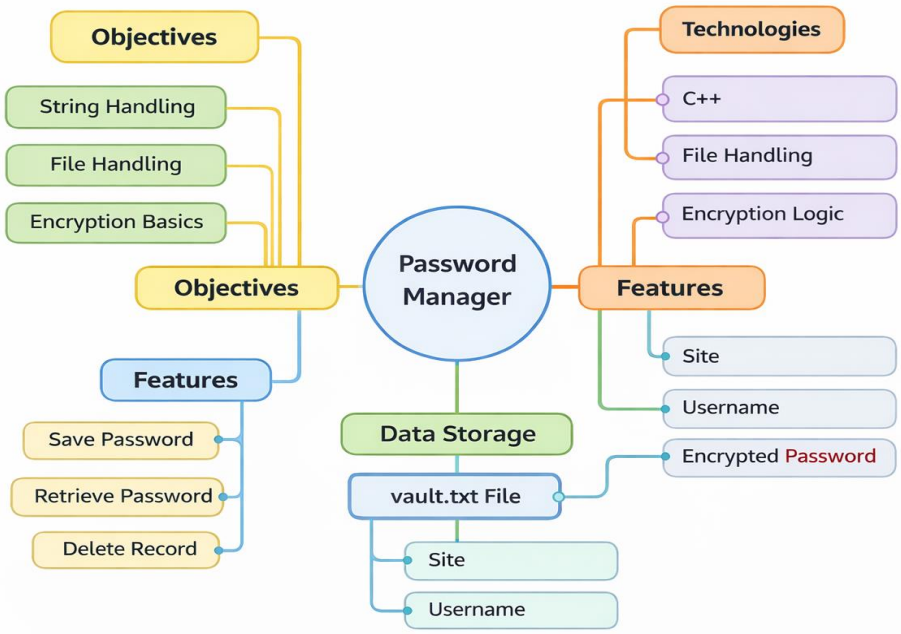
```cpp
#include <fstream>
#include <iostream>
#include <string>
#include <limits>
using namespace std;
```

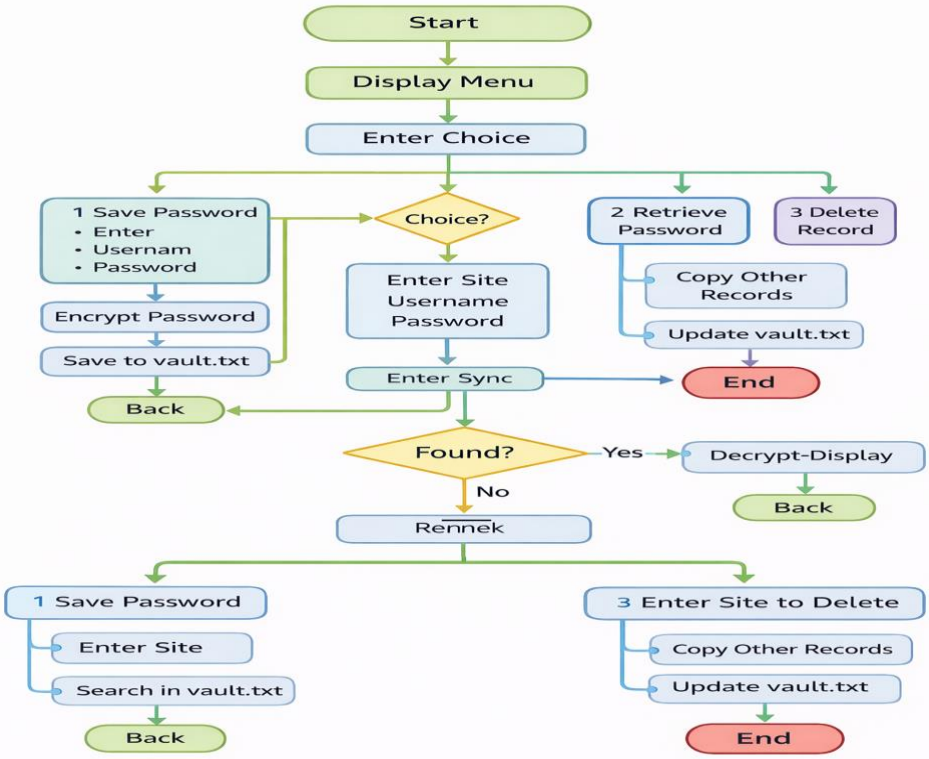
3. **Environment**
    - Any C++ compiler (G++, Clang) and a CLI terminal.
4. **Data Storage**
    - Vault.txt

```
password.cpp
vault.txt
```

5. Problem Statement / Case Study Implementation Details and Snapshots.




Objectives
String Handling
File Handling
Encryption Basics
Objectives
Features
Save Password
Retrieve Password
Delete Record
Password Manager
Data Storage
vault.txt File
Site
Username
Technologies
C++
File Handling
Encryption Logic
Features
Site
Username
Encrypted Password


Start
Display Menu
Enter Choice
1 Save Password
• Enter
• Usernam
• Password
Encrypt Password
Save to vault.txt
Back
Choice?
Enter Site
Username
Password
Enter Sync
Found?
Yes
No
Decrypt-Display
Back
Rennek
2 Retrieve Password
Copy Other Records
Update vault.txt
End
3 Delete Record
1 Save Password
Enter Site
Search in vault.txt
Back
3 Enter Site to Delete
Copy Other Records
Update vault.txt
End

6. Problem Statement / Case Study Results and Conclusion.

Findings

- Using a Shift Cipher, even a basic one, provides a significant security layer over plain-text storage.
- File-based storage is highly efficient for small to medium-scale personal use without the need for complex database servers.
- Pipe-delimited formats allow for handling spaces in site names and usernames effectively.

Conclusion

This case study successfully demonstrates the creation of a functional and secure Password Manager. By integrating file handling, string manipulation, and basic cryptography, the project provides a reliable foundation for personal data security. It proves that CLI tools can be both powerful and accessible for managing critical information in a local environment.

7. References

- C++ Standard Library Documentation: cppreference.com for fstream and string headers.
- Cryptography Basics: Introduction to Caesar and Shift Ciphers in data security.
- **ITM Skills University**: Academic curriculum guidelines for Case Study 9: Password Manager.
- Matplotlib and Seaborn documentation for visualizations.
- GNU Compiler Collection (GCC) Documentation: For compilation and execution standards.